

2026

Towson Data

Analytics Competition

Edward Caraballo

Not All Counties Face The Same Burden

Baltimore County's Uninsured Rate

6%

Kenedy County's Uninsured Rate

38%

Missouri	Dade	5.4	23	6.2	14	1520:1	1543:1	7.6	57	91	3.0	104182	10
Utah	Daggett	4.3	17	5.4	7	976:0	992:0	6.9	61	91	4.2	119650	12
Minnesota	Dakota	3.4	12	4.9	4	1656:1	408:1	9.5	78	95	2.6	185989	11
Nebraska	Dakota	3.8	21	4.4	11	7080:1	394:1	8.0	54	81	2.7	125643	21
Alabama	Dale	5.0	23	5.9	10	3524:1	712:1	6.4	59	86	2.5	103328	12
Texas	Dallam	5.0	25	5.9	28	1195:1	2412:1	6.6	39	81	2.6	103957	8
Alabama	Dallas	5.4	28	6.5	10	1393:1	1292:1	6.4	57	87	5.4	82042	16
Arkansas	Dallas	5.0	25	6.2	9	6308:1	281:1	5.1	60	89	3.9	99531	10
Iowa	Dallas	2.8	11	4.2	4	2359:1	2415:1	9.5	86	96	2.2	190864	8
Missouri	Dallas	5.3	21	6.1	15	17341:1	2538:1	6.0	41	90	3.1	108492	12
Texas	Dallas	4.1	21	5.5	24	1387:1	455:1	7.3	63	82	3.8	147918	21
Wisconsin	Dane	3.4	13	5.1	5	781:1	197:1	9.0	84	96	2.3	168282	14
Montana	Daniels	4.1	16	5.5	12	1686:0	1633:1	8.4	75	97	1.9	111700	11
Virginia	Danville City	5.2	27	6.4	8	898:1	242:1	7.1	47	85	4.6	90959	17
North Carolin	Dare	3.4	12	4.6	11	1719:1	465:1	7.9	71	95	4.0	154906	14
Ohio	Darke	4.7	19	6.1	7	2150:1	1028:1	8.1	54	91	3.1	116477	9
South Carolin	Darlington	4.4	20	5.7	11	1846:1	630:1	7.6	57	87	3.3	108258	13
Pennsylvania	Dauphin	4.1	19	5.5	7	968:1	346:1	8.1	67	91	3.2	139938	14
North Carolin	Davidson	4.1	18	5.1	12	3555:1	791:1	7.1	60	88	3.4	118595	11
Tennessee	Davidson	4.2	20	6.1	13	1105:1	230:1	7.8	75	90	2.7	149681	17
North Carolin	Davie	3.9	17	4.9	10	2419:1	970:1	7.4	63	91	3.1	139303	13
Indiana	Daviess	4.8	22	5.9	17	2386:1	1086:1	8.3	47	77	2.4	127061	13
Kentucky	Daviess	4.5	20	5.3	6	1808:1	309:1	7.7	63	91	4.3	122867	14
Missouri	Daviess	5.1	23	5.9	16	8399:1	4276:1	7.7	47	86	2.8	113028	12
Iowa	Davis	4.1	18	5.1	14	4569:1	9169:1	9.1	56	80	2.4	127653	10
Utah	Davis	4.4	14	5.6	6	2186:1	369:1	8.6	76	96	2.4	180325	9
South Dakota	Davison	3.6	16	4.6	11	1242:1	623:1	7.6	63	92	1.7	126110	11
Nebraska	Dawes	3.7	15	4.6	8	1358:1	290:1	7.5	83	96	2.0	110630	13
Georgia	Dawson	4.0	15	5.1	14	1500:1	992:1	8.7	73	89	2.6	166778	10
Montana	Dawson	3.7	16	5.2	8	2226:1	383:1	8.0	75	94	2.6	122727	10
Nebraska	Dawson	4.1	20	4.4	13	1838:1	602:1	6.7	48	78	2.4	117049	15
Texas	Dawson	5.5	30	6.0	22	2483:1	12004:1	5.0	44	72	4.2	106672	18
South Dakota	Day	3.6	15	4.4	13	2707:1		6.8	66	92	2.7	121972	9
New Mexico	De Baca	4.9	21	6.1	13	1680:1	87:1	7.5	70	88	3.3	112791	7
Louisiana	De Soto	5.0	23	6.6	8	5384:1	411:1	6.8	55	85	4.1	123309	11

The Approach

1

Rank

Every county across multiple health metrics

2

Find Overlap

Between high uninsured rates and other poor health outcomes

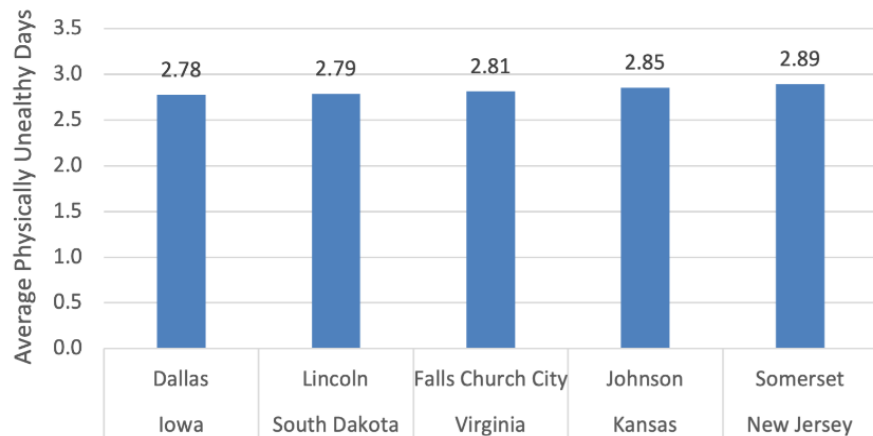
3

Identify Predictors

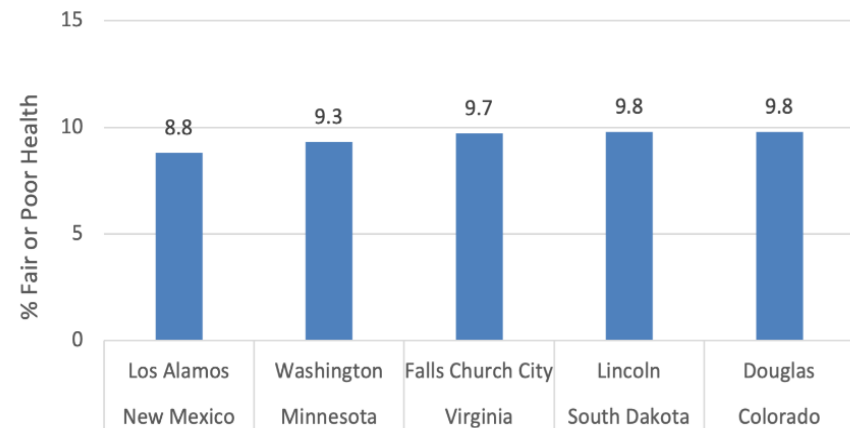
Run regression analysis to find the strongest predictors of poor health outcomes

Best Performing Counties

Top 5 Counties with Lowest Average Number of Physically Unhealthy Days

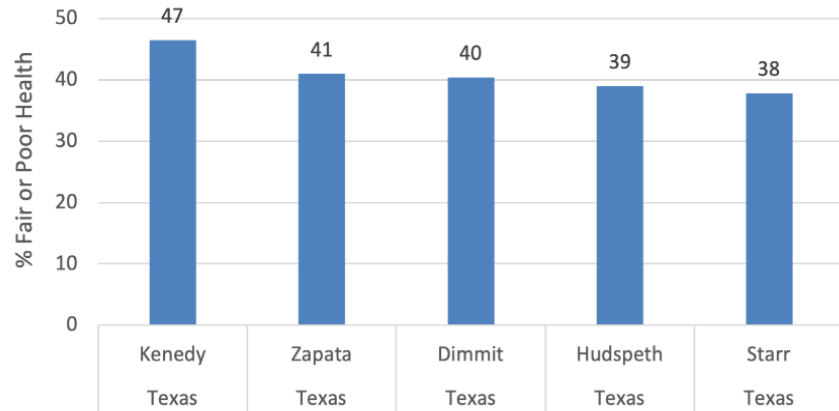


Top 5 Counties with the Lowest Percentage of Fair or Poor Health

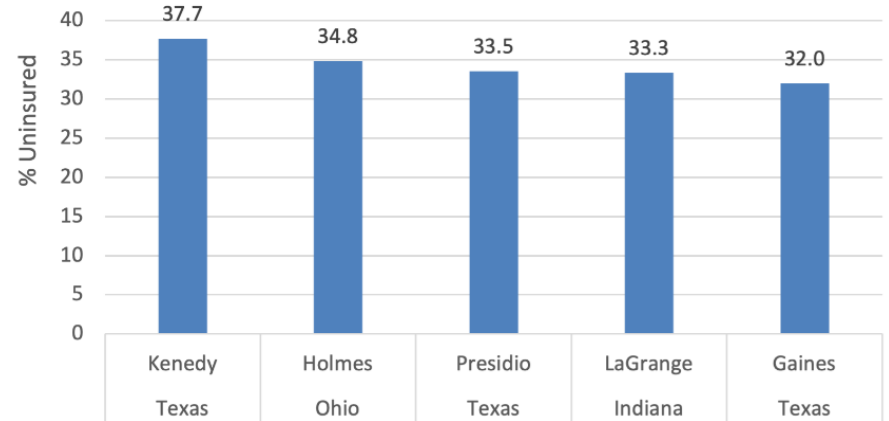


Worst Performing Counties

Top 5 Counties with the Highest Percentage of Fair or Poor Health



Top 5 Counties with the Highest Percentage Uninsured



Counties with Highest Uninsured Rates and Overlapping Health Vulnerabilities

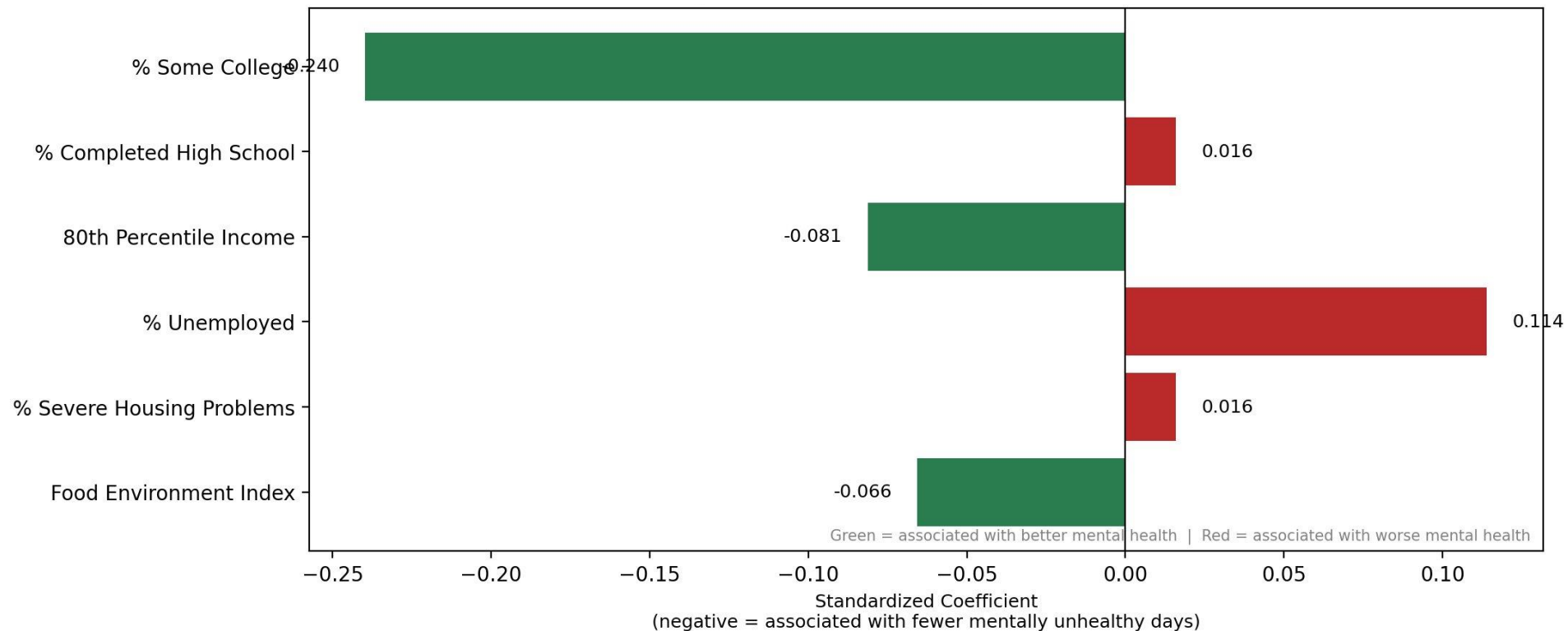
Lower rank = worse outcome · Sorted by number of outcomes in worst 150 nationally

County	Uninsured rank	Poor health rank	Phys. unhealthy days rank	Mental unhealthy days rank	Outcomes in worst 150
Kenedy, Texas	#1	1	2	84	● ● ●
Presidio, Texas	#3	6	83	921	● ● ●
Hidalgo, Texas	#6	8	108	925	● ● ●
Edwards, Texas	#7	59	169	700	● ● ●
Holmes, Ohio	#2	395	495	226	● ● ●
Swisher, Texas	#9	223	353	983	● ● ●
Gaines, Texas	#5	512	531	824	● ● ●
Dallam, Texas	#10	407	663	1015	● ● ●
LaGrange, Indiana	#4	782	971	532	● ● ●
Ochiltree, Texas	#8	456	890	1400	● ● ●

< 100 (worst)
100-400
400-700
700-1000
> 1000 (best)
● ● ● = 2 of 3 outcomes in worst 150

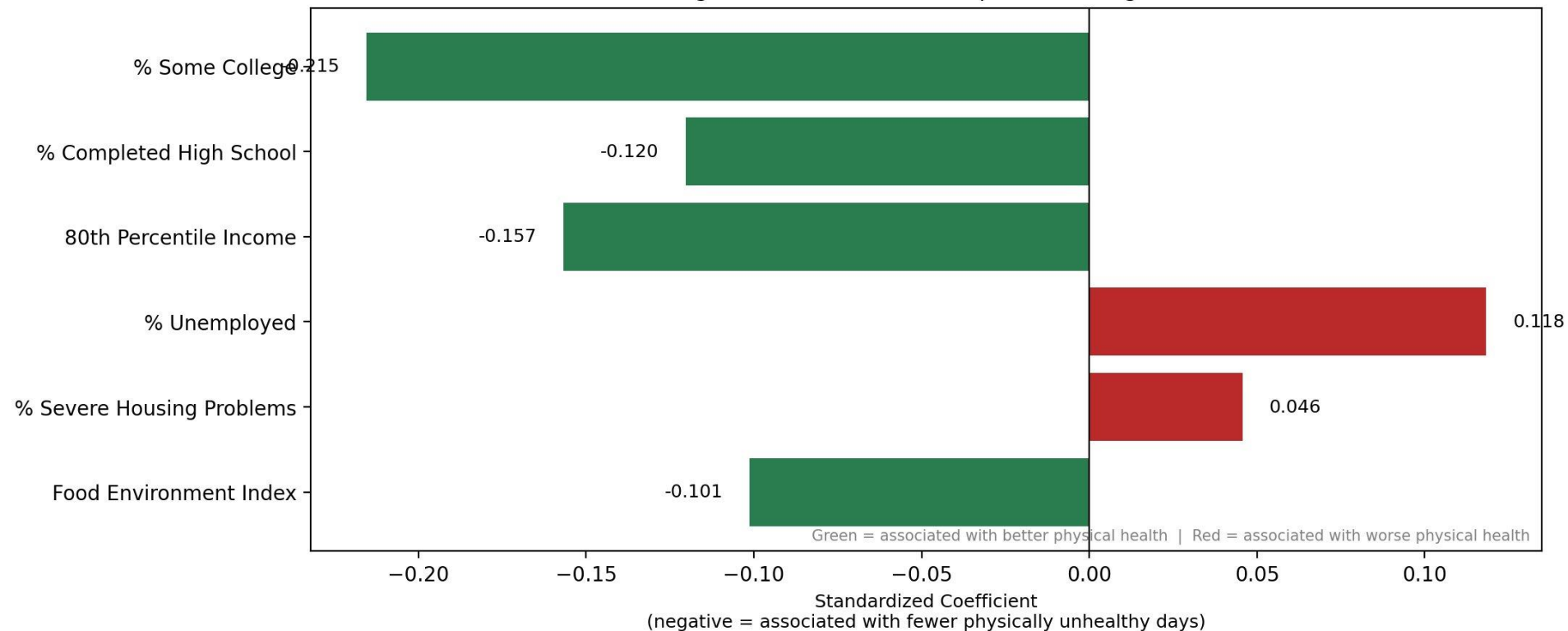
What Predicts Mental Health?

Strongest Predictors from Multiple Linear Regression



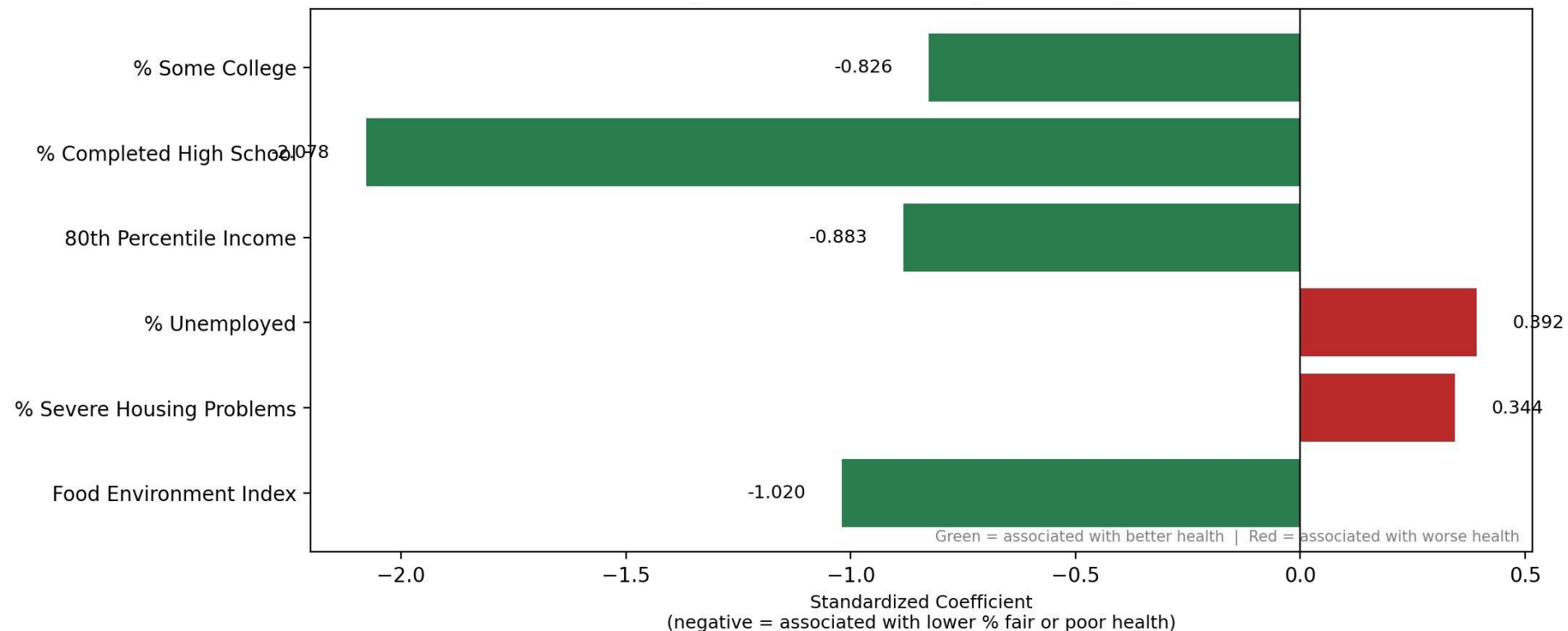
What Predicts Physical Health?

Strongest Predictors from Multiple Linear Regression



What Predicts Fair or Poor Health?

Strongest Predictors from Multiple Linear Regression



Short-Term Recommendation

Bottom 7%

nationally for unemployment rate
(Kenedy, Presidio & Hidalgo)

63%

of working-age Americans get
insurance through their employer

The health department can't fix unemployment — but it can reduce the health consequences.

Enrollment Outreach

Help residents who qualify for
Medicaid or Marketplace coverage
actually know it and sign up

Mobile Clinics

Increase access in parts of these
counties where getting care may be
difficult

Community Health Workers

Help residents navigate coverage,
connect to local care, and stay on top
of preventive services

Education Is Still The Strongest Predictor

Long-term, the Health Department should seek partnerships with:

School Districts

**Community
Colleges**

ACTIONABLE RECOMMENDATION

Support high school programs that help students prepare for college and navigate the application process

The Clearest Conclusion Is:

**Kenedy County,
TX**

**Presidio County ,
TX**

**Hidalgo County ,
TX**

Should be prioritized for targeted intervention

Short Term

Insurance and Care Access

Long Term

Broader Educational
Opportunity